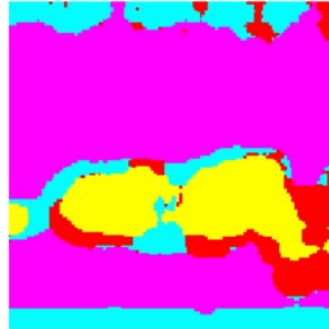


Chapter 5 – Extraction of color and texture

Visual Computing



(a) Original image



(b) Segmentation into 4 clusters



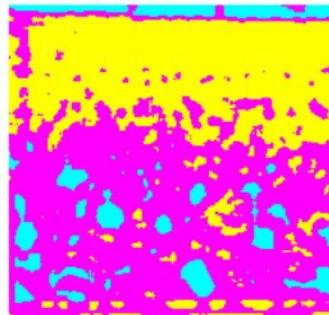
(c) Original image



(d) Segmentation into 4 clusters



(e) Original image



(f) Segmentation into 3 clusters



Many images obtained with CCD are in color.

This issue raises the following questions -> *What is the color ?*
how can we represent it in a PC ?

Definition:

The color is the **human perception**, that is, the output of the visual system to the electromagnetic waves (i.e., light) arriving at the retina cells

Facts:

- In the 19th century, Young e Helmholtz found that it is possible synthesize many colors (not all !) from the mixture of 3 primary colors.
- This is a remarkable discovery, since it allows the reproduction of colors in both PC and TV

- Suppose that we receive a light beam at the retina with the spectrum $S(\lambda)$

where λ is the wave length

- Also, suppose that the primary colors have a known spectrum, i.e.,
 $P_j(\lambda), j = 1, 2, 3$

In general $S(\lambda)$ is a complex function and can not be represented by the linear combination of the three spectra $P_j(\lambda)$ that is

$$\overset{1}{S(\lambda)} \neq \overset{2}{\sum_{j=1}^3 c_j P_j(\lambda)}$$

(where c_j are the mixture coefficients)

Question:

How can we relate these facts with the Young e Helmholtz's discovery ?

Although, the three colors do not suffice to represent the input spectrum $S(\lambda)$ what is amazing is that, despite this difference, the color perception can be the same !!

To better understand this, we have to figure out how the retina works !

- The retina has three types of cells that are sensitive to color – the cones !
- Two spectra may be different and still produce the same output in the three types of cones
- The output of each cone $i \in \{1, 2, 3\}$ to the incident spectrum is given by

$$r_i = \int S(\lambda) S_i(\lambda) d\lambda$$

(where $S_i(\lambda)$ is the output of cone i)

Using the equation

$$r_i = \int S(\lambda) S_i(\lambda) d\lambda$$

We can conclude that ¹ $S(\lambda)$ and ² $\sum c_j P_j(\lambda)$ represent the **same** color if and only if

$$\int S(\lambda) S_i(\lambda) d\lambda = \int \sum_j c_j P_j(\lambda) S_i(\lambda) d\lambda \quad i = 1, 2, 3$$

which is equivalent to

$$\int S(\lambda) S_i(\lambda) d\lambda = \sum_j c_j \int P_j(\lambda) S_i(\lambda) d\lambda \quad i = 1, 2, 3$$

Defining

$$R_{ij} = \int P_j(\lambda) S_i(\lambda) d\lambda$$
$$r_i = \int S(\lambda) S_i(\lambda) d\lambda$$

We can write

$$\mathbf{r} = \mathbf{R}\mathbf{c}$$

This is the condition which must be met in order to obtain the **same color perception** with three primary colors !

There exists some constraints (i.e., some conditions must be met in order to **obtain the same color perception** with three primary colors):

- The mixture is realizable if the coefficients are $c_i \geq 0$ (it is not true in some cases)
- When the solution leads to a negative coefficients, this means that the color is not realizable by the mixture of three primary colors
- *Can three primary spectra $P_j(\lambda)$ synthesize all the colors with positive coefficients ?*

The answer is **No**!

Color – several coordinate systems

Color can be represented by several systems of coordinates:

- **RGB** : 700 nm (red), 546.1 nm (green) 435.8 nm (blue) – the most popular choice
- **XYZ**
- **YUV**
- **HSV** (hue, saturation, value)
- **CMY** (used for printing)



Color images



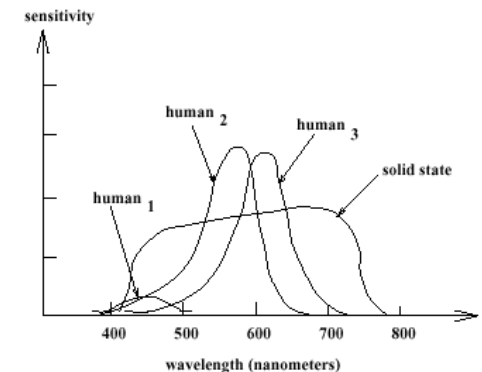
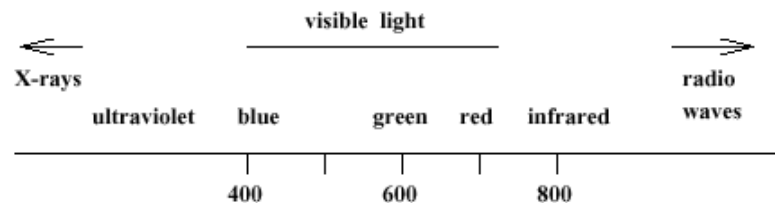
RGB



HSV

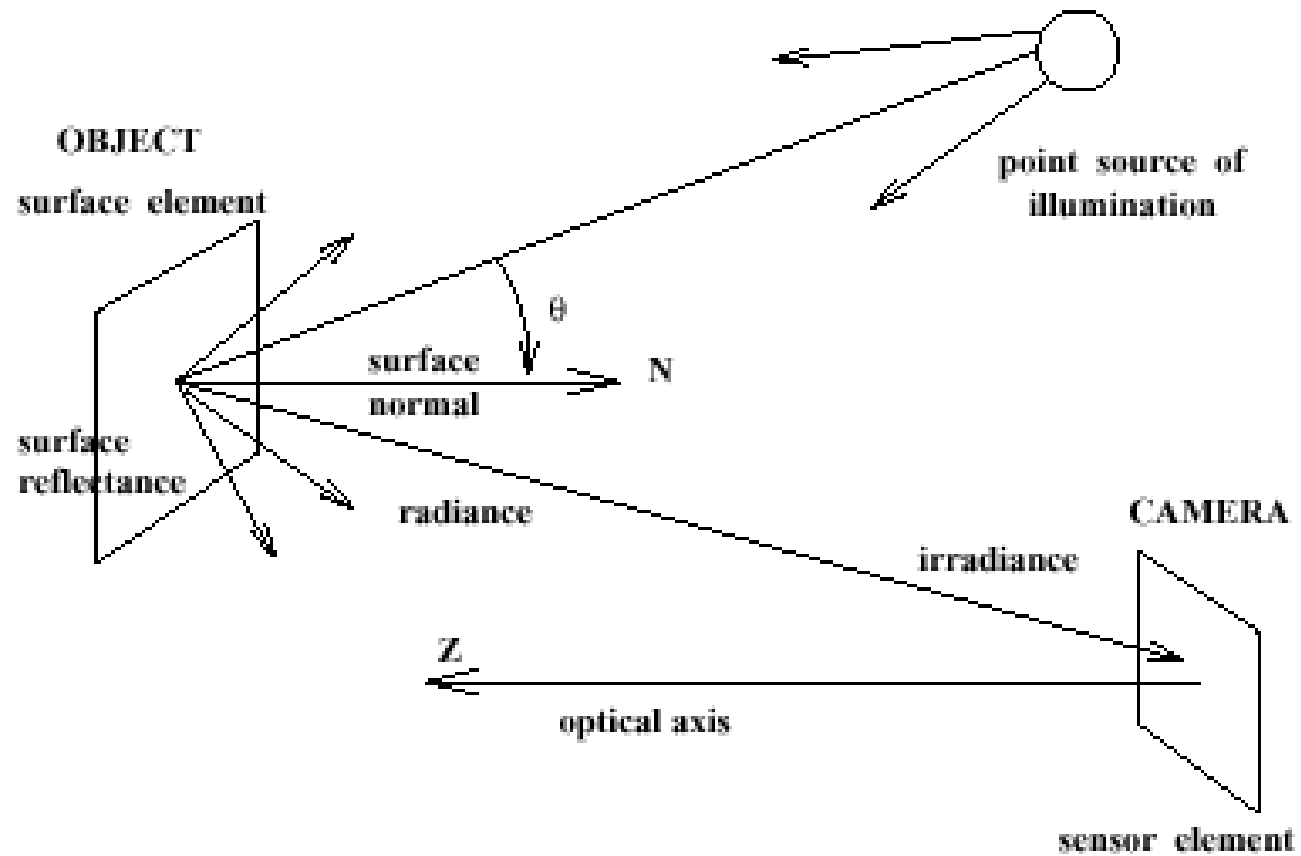
Color Perception

- **Human perception** depends on 3 main factors:
 - The way how the light source distributes in the spectra
 - The reflectance of the object surface, that is, the relation between the emission spectrum and the source spectrum radiated from the surface
 - The spectra sensibility of the sensor
 - “An object is ‘blue’ if illuminated with white color - looks like- blue”.
 - “The same object turns violet, if illuminated with red color”. “A blue car illuminated with intense sunlight (white) heats up and irradiates energy in the IR band” (invisible for human eye, but visible be IR sensor)



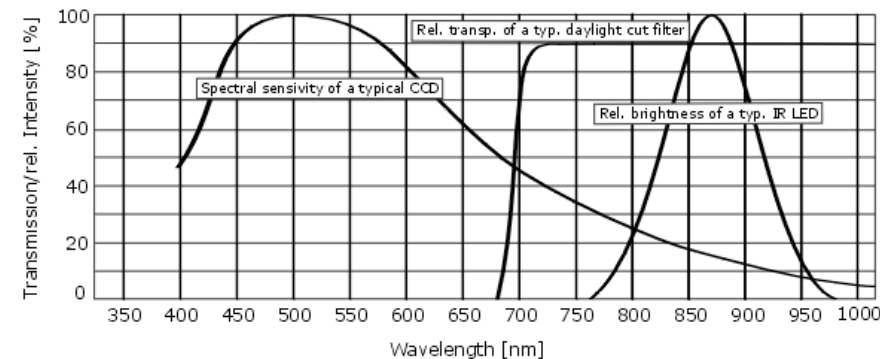
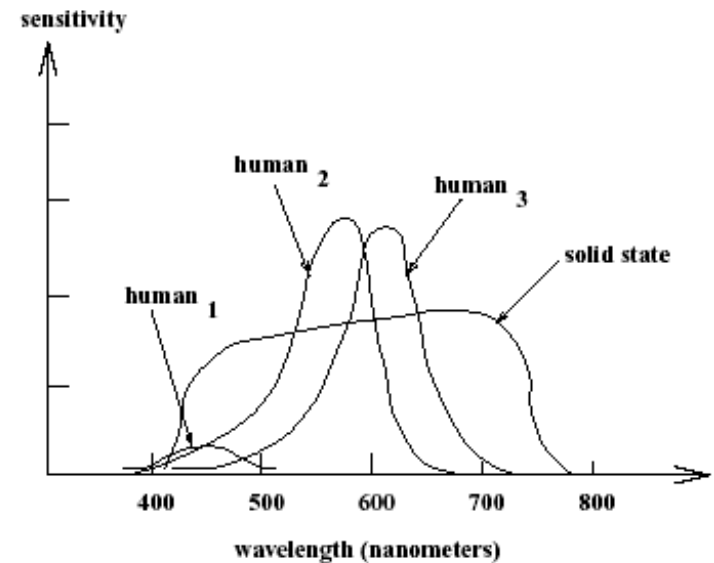
- There are another issues that affect the object perception:
 - Material (specular surfaces), distance, orientation

Color Perception



Sensor Sensibility

- The retina color receptors (*cones*) are sensible only in a given range of the wave-length
- The Human Visual System (HVS) has three types of *cones*
- The brain is responsible for the fusion of the information of these 3 fonts – Perception and color
- How is that possible?
 - There exists infinite possibilities of the spectra distribution. However, only three characteristics are necessary
- Important note:
 - The CCD sensors have, in general, good sensibility in the IR band (advantage or disadvantage ?)



Summary

- The perception of the color depends on:

- Light source

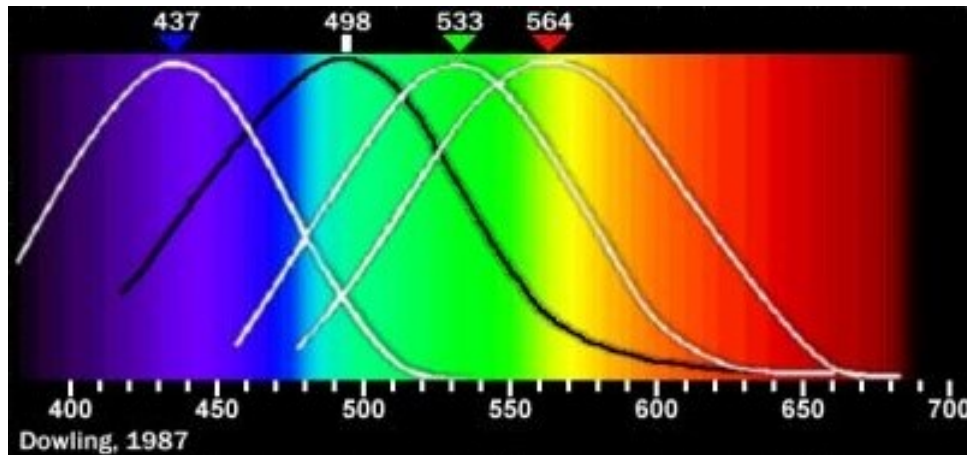
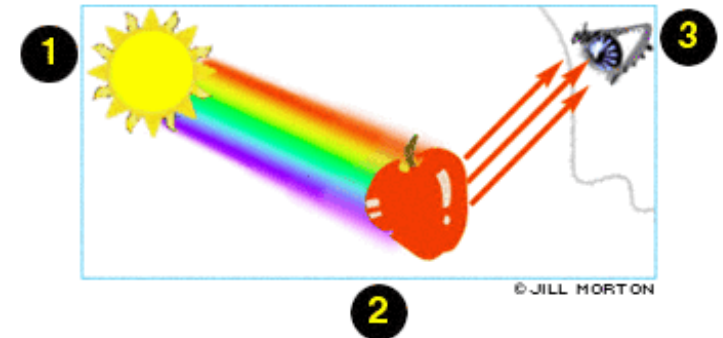
$$E(\lambda)$$

- Object reflectance (*albedo*)

$$S(\lambda)$$

- Observer sensibility

$$f_C(\lambda), C \in \{R, G, B\}$$



(dark line → sensibility of the *rods*)

$$G = \int_{\lambda} E(\lambda) S(\lambda) f_G(\lambda) d\lambda$$

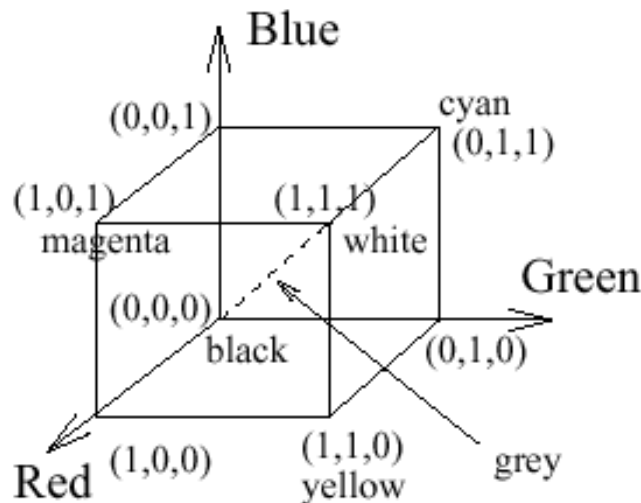
$$B = \int_{\lambda} E(\lambda) S(\lambda) f_B(\lambda) d\lambda$$

$$R = \int_{\lambda} E(\lambda) S(\lambda) f_R(\lambda) d\lambda$$

Color representation in RGB

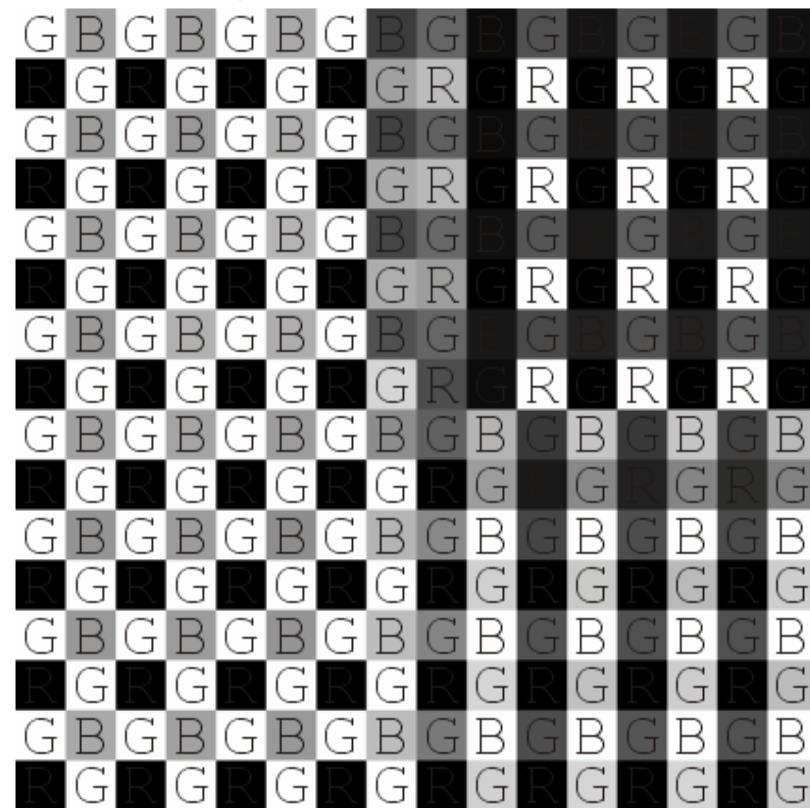
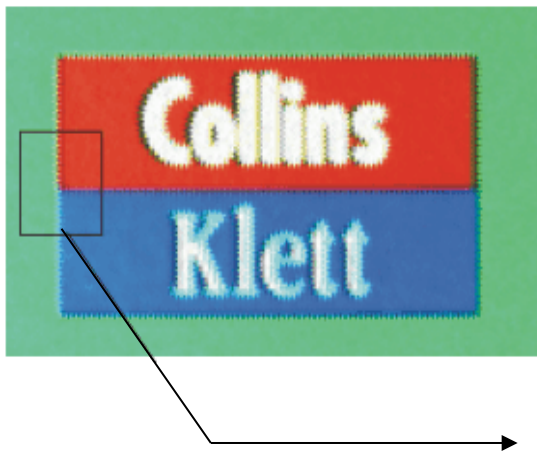
- Currently, the graphical systems use 3 bytes (RGB) for representing the color of a pixel (*true color*) $\rightarrow$ 16.777.216 possible codifications
 - 16 bits/pixel is a reasonable choice (5 bits for each of the components RGB, plus one additional bit for green). The HVS has larger sensibility in the green band.

RGB cube

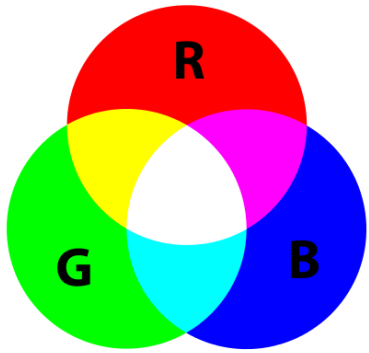


Color representation in RGB

Monitor RGB



Other Color Representation Systems



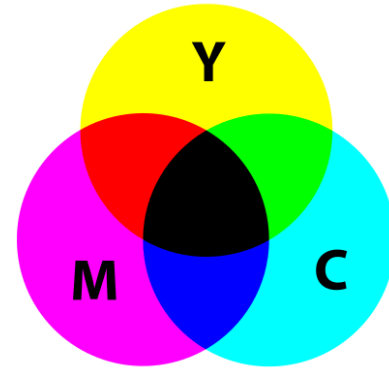
Additive
system

RGB

Subtractive
system

CMY

HSI



RED	(255, 0, 0)	(0, 255, 255)	(0.0 , 1.0, 255)
YELLOW	(255, 255, 0)	(0, 0, 255)	(1.05, 1.0, 255)
	(100, 100, 50)	(155, 155, 205)	(1.05, 0.5, 100)
GREEN	(0, 255, 0)	(255, 0, 255)	(2.09, 1.0, 255)
BLUE	(0, 0, 255)	(255, 255, 0)	(4.19, 1.0, 255)
WHITE	(255, 255, 255)	(0, 0, 0)	(-1.0, 0.0, 255)
GREY	(192, 192, 192)	(63, 63, 63)	(-1.0, 0.0, 192)
	(127, 127, 127)	(128, 128, 128)	(-1.0, 0.0, 127)
	(63, 63, 63)	(192, 192, 192)	(-1.0, 0.0, 63)
	...		
BLACK	(0, 0, 0)	(255, 255, 255)	(-1.0, 0.0, 0)

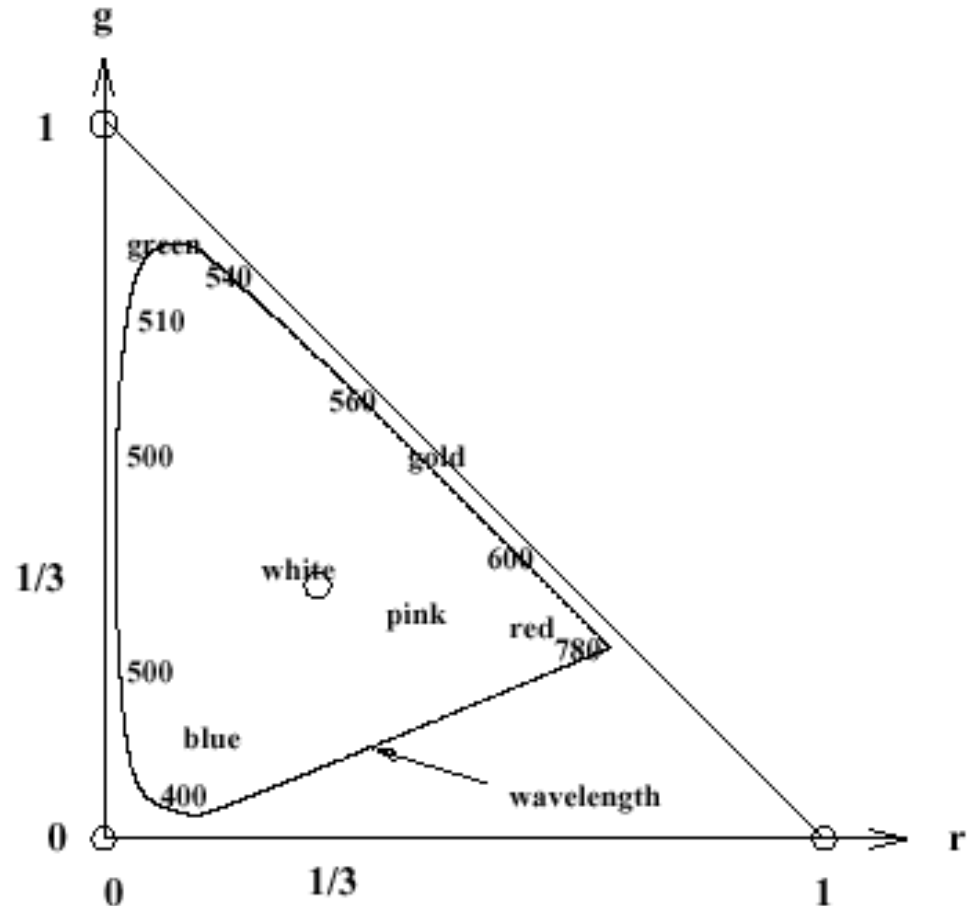
15

$$g = \frac{G}{R + G + B}$$

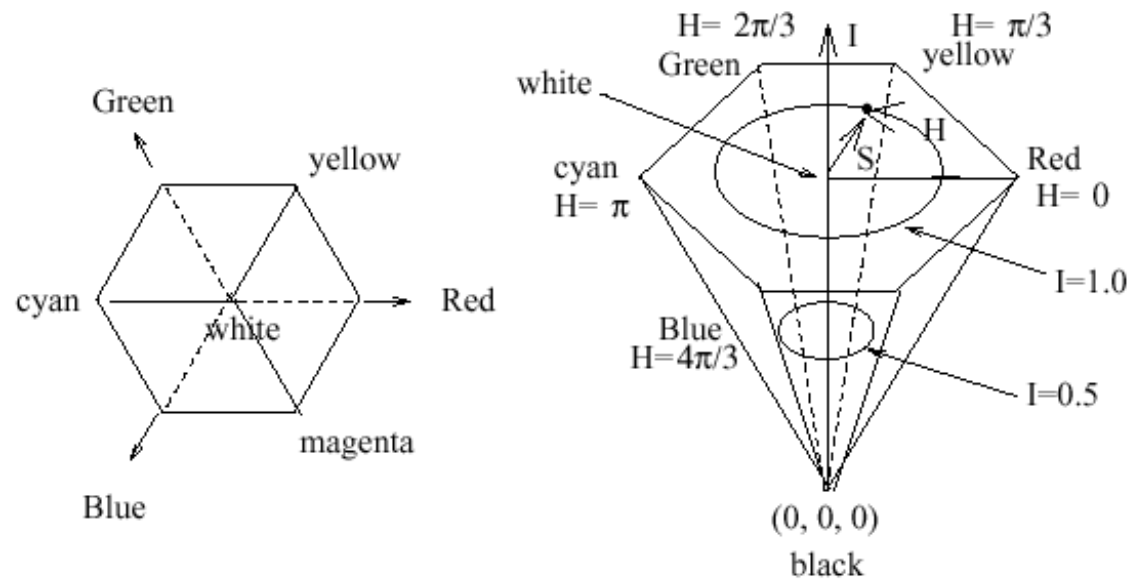
$$I = (R + G + B)/3$$

Other alternative:

- normalization by $\max(R,G,B)$



Hue Saturation Intensity (HSI) representation



Resulting effect by changing the saturation component



original

+40%

-20%

Conversion from RGB to HSI

Conversion of RGB encoding to HSI encoding.

R,G,B : input values of RGB all in range [0,1] or [0,255];

I: output value of intensity in same range as input;

S : output value of saturation in range [0,1];

H : output value of hue in range [0,2 π), -1 if S is 0;

R,G,B,H,S,I are all floating point numbers;

```
procedure RGB_to_HSI( in R,G,B; out H,S,I)
{
  I := max ( R, G, B );
  min := min ( R, G, B );
  if (I  $\geq$  0.0) then S := (I - min )/I else S := 0.0;
  if (S  $\leq$  0.0) then { H := -1.0; return; }
  “compute the hue based on the relative sizes of the RGB components”
  diff := I - min;
  “is the point within +/- 60 degrees of the red axis?”
  if (r = I) then H := ( $\pi/3$ )*(g - b)/diff;
  “is the point within +/- 60 degrees of the green axis?”
  else if (g = I) then H := (2 *  $\pi/3$ ) +  $\pi/3$  *(b - r)/diff;
  “is the point within +/- 60 degrees of the blue axis?”
  else if (b = I) then H := (4 *  $\pi/3$ ) +  $\pi/3$  *(r - g)/diff;
  if (H  $\leq$  0.0) H := H + 2 $\pi$ ;
}
```

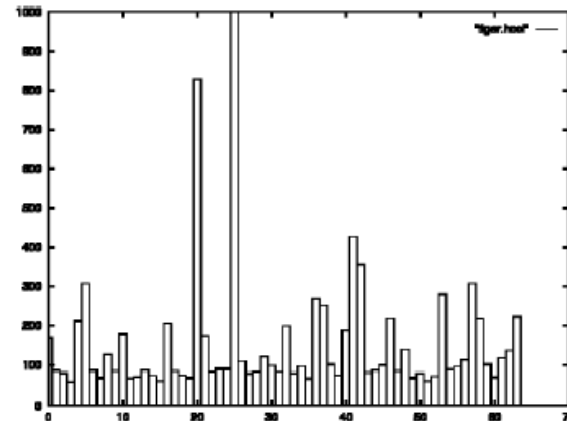
$$V = \max(R, G, B)$$

$$S = \frac{255 (V - \min(R, G, B))}{V} \quad \text{if } V \neq 0, 0 \text{ otherwise}$$

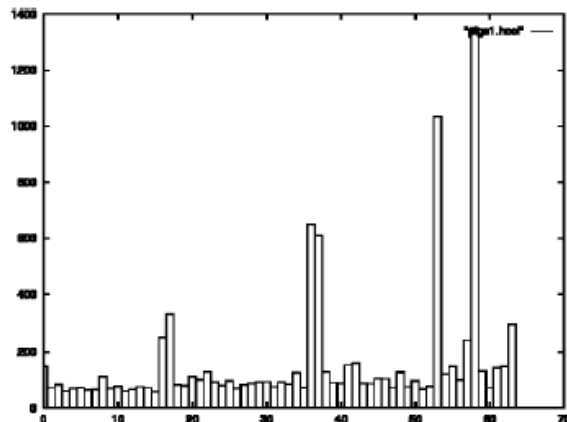
$$H = \begin{cases} \frac{60 (G - B)}{S} & \text{if } V = R \\ 120 + \frac{60 (B - R)}{S} & \text{if } V = G \\ 240 + \frac{60 (R - G)}{S} & \text{if } V = B \end{cases}$$

$$\text{if } H < 0 \text{ then } H = H + 360$$

Color Histogram

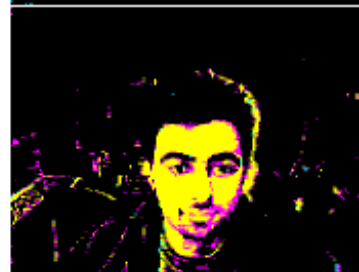


$$Intersection\{h_I, h_M\} = \sum_{j=1}^K \min\{h_I(j), h_M(j)\}$$



$$match\{h_I, h_M\} = \frac{\sum_{j=1}^K \min\{h_I(j), h_M(j)\}}{\sum_{j=1}^K h_M(j)}$$

Classification using Color



Classification (face detection - I)

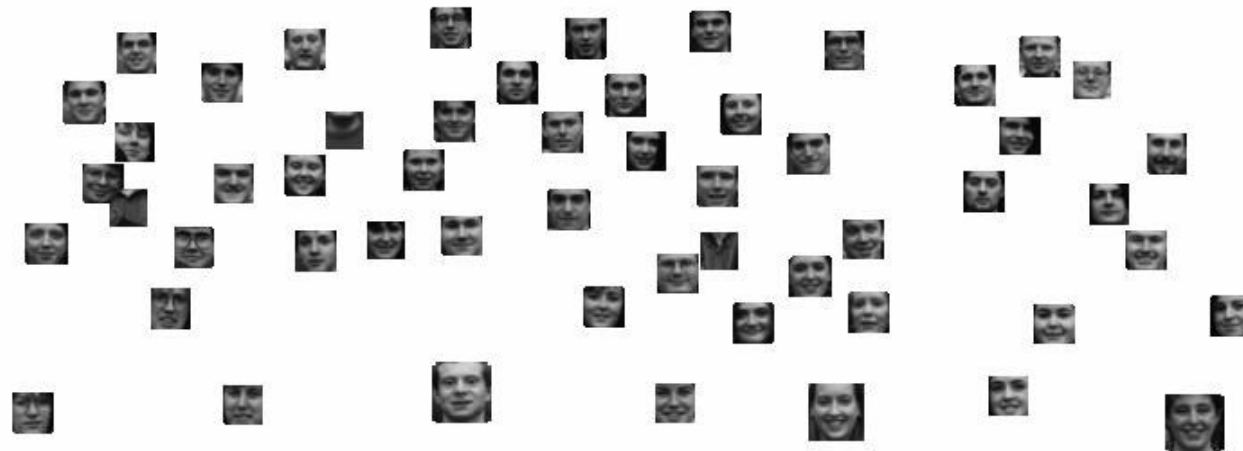


Classification (face detection - IV)

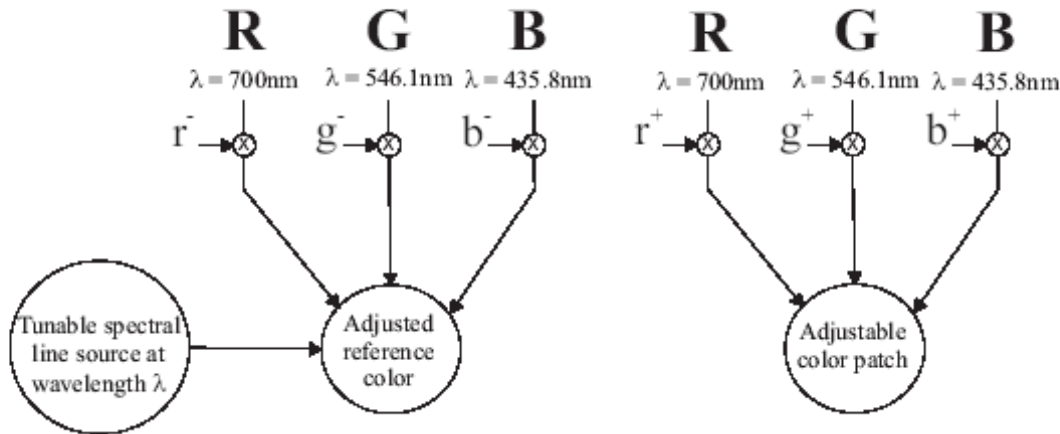


Classificação (detecção de faces - III)





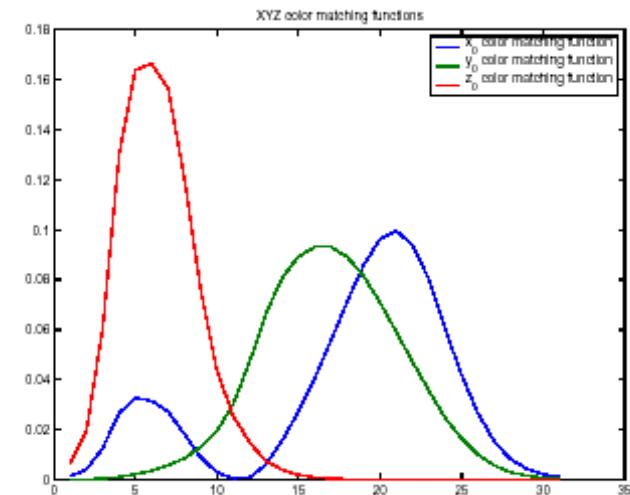
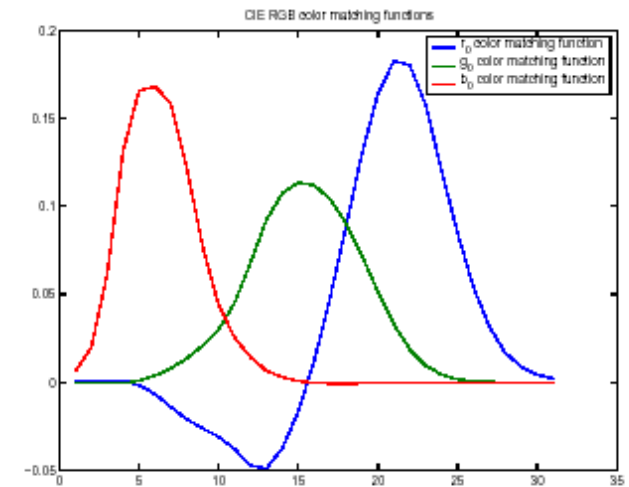
Primary colors: R, G, B



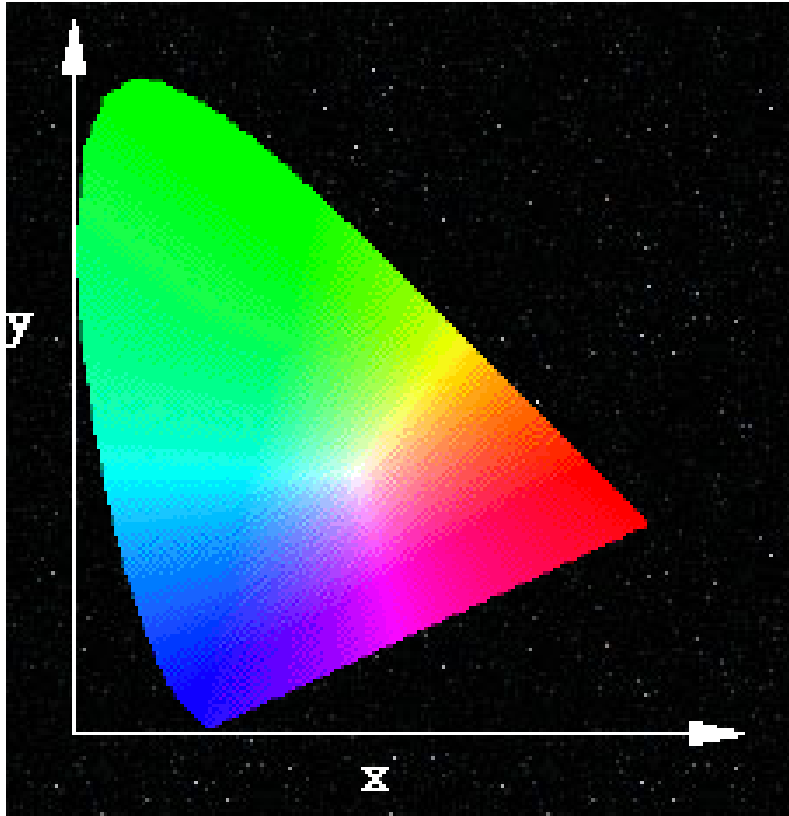
Problem: Some colors produce negative coefficients

Solution: Linear transform.

Primary colors are now imaginárias $\rightarrow$ XYZ



Chromaticity Diagram x,y



Chromatic coordinates

$$x = \frac{X}{X + Y + Z}$$

$$y = \frac{Y}{X + Y + Z}$$

$$z = 1 - x - y$$

Relation with primary colors, RGB:

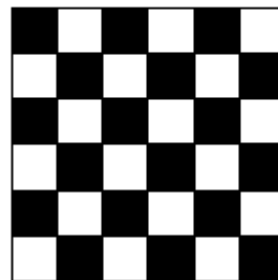
$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} 0.619 & 0.177 & 0.204 \\ 0.299 & 0.586 & 0.115 \\ 0.000 & 0.056 & 0.944 \end{pmatrix} \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

What is the texture?

- It is hard to define the texture meaning
- Texture give us the information about the spatial distribution of the intensities and/or colors
- It is a useful feature for segmenting images in regions
- Example:
 - Different textures with the same histogram



block pattern

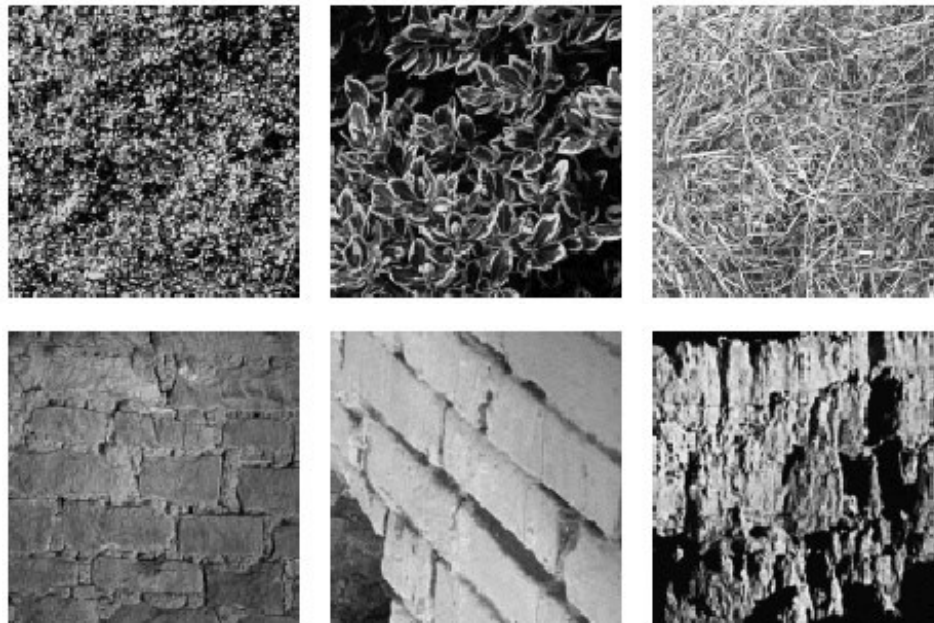


checkerboard



striped pattern

- Structural Approach
 - texture is the way how the set of basic patterns (*texels*) are organized in a region
- Statistical Approach
 - texture is a quantitative measure of how the intensities are arranged in a region



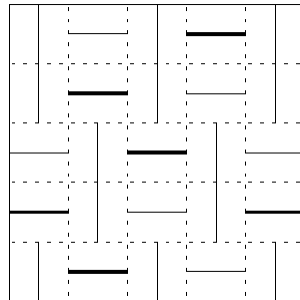
- Density (*edgeness*) and edges orientation

$$F_{edgeness} = \frac{|\{p \mid \text{Mag}(p) \geq T\}|}{N}$$

$$F_{magdir} = (H_{mag}(R), H_{dir}(R))$$

– Example 1

- 2 levels of amplitude: (weak , strong)
- 3 levels of orientation: (horizontal , vertical , diagonal)



$$F_{edgeness} = \frac{25}{25} = 1$$

$$F_{magdir} = ((0.24, 0.76)(0.48, 0.52, 0.00))$$

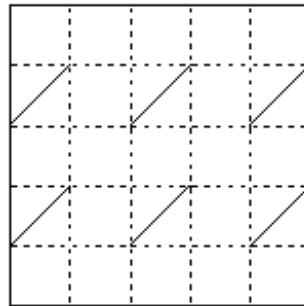
- Density (*edgeness*) and edges orientation

$$F_{edgeness} = \frac{|\{p \mid \text{Mag}(p) \geq T\}|}{N}$$

$$F_{magdir} = (H_{mag}(R), H_{dir}(R))$$

– Example 2

- 2 levels of amplitude: (weak , strong)
- 3 levels of orientation: (horizontal , vertical , diagonal)



$$F_{edgeness} = \frac{6}{25} = 0.24$$

$$F_{magdir} = ((0.00, 0.24)(0.00, 0.00, 0.24))$$

- Histogram distance $L_1(H_1, H_2) = \sum_{i=1}^n |H_1(i) - H_2(i)|$

2nd Order Gray level statistics

- **1st order statistics are limited** – do not account for spatial distribution of gray levels in the image
- **2nd order statistics can overcome this limitation**, since they account for the spatial distribution of the gray levels in the image
- It is possible to describe such spatial distribution, using the so called **Gray Level Co – Occurrence Matrix (GLCM)**

Gray Level Co – Occurrence Matrix

- GLCM is based on the 2nd order probability density function $p(i, j | d, \theta)$, that can be estimated for several distances d and directions θ ;
- Thus, $p(i, j | d, \theta)$ can be described as the probability of two co-linear pixels, with a given direction θ and with the distance d , of having a gray level in i and j .
- The functions $p(i, j | d, \theta)$ can be described in a matrix formulation given by the equation

$$\Omega(d, \theta) = p(i, j | d, \theta), \quad 0 \leq i, j < L$$

Gray Level Co – Occurrence Matrix. Examples

- These matrices are of size $L \times L$, being L the maximum gray level of the ROI and it is symmetric, since $p(i,j | d, \theta) = p(j,i | d, \theta)$.
- The analysis of this matrix allows to better comprehend the region texture
- Examples for the calculation of matrix $\Omega(d, \theta)$ given the directions $\theta = \{0^\circ, 45^\circ\}$ and for $d=1$

0	0	1	1
1	2	1	3
1	0	3	0
2	0	3	1

$I(x, y)$

	0	1	2	3
0	2/24	2/24	1/24	3/24
1	2/24	2/24	2/24	2/24
2	1/24	2/24	0/24	0/24
3	3/24	2/24	0/24	0/24

$\Omega(1, 0^\circ)$

	0	1	2	3
0	0/18	2/18	1/18	2/18
1	2/18	2/18	2/18	0/18
2	1/18	2/18	0/18	0/18
3	2/18	0/18	0/18	2/18

$\Omega(1, 45^\circ)$